(1)

a) First, I will derive the solution that is optimal, then show that it's unique. We have:

$x_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$, $w \in \mathbb{R}^d$, $N \geq d$, $X = [x_1, \dots, x_N]^T \in \mathbb{R}^{N \times d}$, $y = [y_1, \dots, y_N] \in \mathbb{R}^N$, $X^T X$ invertible. Then:

$$L(w) = \frac{1}{2N} \sum_{i=1}^{N} (y_i - w^T x_i)^2 =$$

$$= \frac{1}{2N} \sum_{i=1}^{N} y_i^2 - 2 y_i (w^T x_i) + (w^T x_i)^2 .$$

First, L is convex and differentiable: Let $\alpha \in (0,1)$, and $w_1, w_2 \in \mathbb{R}^d$.

Then: $L(\alpha w_1 + (1-\alpha) w_2) = \frac{1}{2N} \sum_{i=1}^{N} y_i^2 - 2 y_i \left((\alpha w_1 + (1-\alpha) w_2)^T x_i \right) +$

$+ \underbrace{\alpha^2 (w_1^T x_i)^2 + 2\alpha(1-\alpha)(w_1^T x_i)(w_2^T x_i) + (1-\alpha)^2 (w_2^T x_i)^2}$

$\leq \alpha (w_1^T x_i)^2 + (1-\alpha)(w_2^T x_i)^2$, Pf: $\alpha (w_1^T x_i)^2 + (1-\alpha)(w_2^T x_i)^2 - \alpha^2 (w_1^T x_i^2) - (1-\alpha)^2 (w_2^T x_i)^2$

$- 2\alpha(1-\alpha)(w_1^T x_i)(w_2^T x_i) = (\alpha - \alpha^2)(w_1^T x_i)^2 + (1 - \alpha - (1-\alpha)^2)(w_2^T x_i^2) +$

$- 2\alpha(1-\alpha)(w_1^T x_i)(w_2^T x_i) = \alpha(1-\alpha)(w_1^T x_i)^2 + \alpha(1-\alpha)(w_2^T x_i)^2 - 2\alpha(1-\alpha)(w_1^T x_i)(w_2^T x_i)$

$= \underbrace{\alpha(1-\alpha)}_{\geq 0} \underbrace{(w_1^T x_i - w_2^T x_i)^2}_{\geq 0} \geq 0.$

So: $L(\alpha w_1 + (1-\alpha) w_2) = \frac{1}{2N} \sum_{i=1}^{N} \alpha y_i^2 + (1-\alpha) y_i^2 - 2 y_i \alpha w_1^T x_i - 2 y_i (1-\alpha) w_2^T x_i$

$+ \alpha^2 (w_1^T x_i)^2 + 2\alpha(1-\alpha)(w_1^T x_i)(w_2^T x_i) + (1-\alpha)^2 (w_2^T x_i)^2 \leq \frac{1}{2N} \sum_{i=1}^{N} \alpha y_i^2 + (1-\alpha) y_i^2$

$- 2 y_i \alpha w_1^T x_i - 2 y_i (1-\alpha) w_2^T x_i + \alpha (w_1^T x_i)^2 + (1-\alpha)(w_2^T x_i)^2 =$

$= \frac{1}{2N} \alpha \sum_{i=1}^{N} y_i^2 - 2 y_i w_1^T x_i + (w_1^T x_i)^2 + \frac{1}{2N} (1-\alpha) \sum_{i=1}^{N} y_i^2 - 2 y_i w_2^T x_i + (w_2^T x_i)^2 =$

$= \alpha L(w_1) + (1-\alpha) L(w_2)$, so $L(w)$ is convex in w □.

Since L is convex, from it's properties, any local minimum of L is a global minimum, and the global minimum is unique.

So that shows that the optimal solution ($\min_w L(w)$) is unique.

Then:

$$\nabla_w L(w) = \frac{1}{2N} \sum_{i=1}^{N} 0 - 2 y_i \underbrace{x_i}_{\nabla_w (w^T x_i)} + \underbrace{2 (w^T x_i) x_i}_{\nabla_w (w^T x_i)^2}$$

$\Rightarrow$ For $w^* = \min_w L(w)$, $\nabla L(w^*) = 0$, so:

$$\frac{1}{2N} \sum_{i=1}^{N} (-2 y_i x_i + 2 (w^{*T} x_i) x_i) = 0$$

$\Leftrightarrow$

$$\sum_{i=1}^{N} \underbrace{(w^{*T} x_i) x_i}_{} = \sum_{i=1}^{N} y_i x_i = X^T y$$

$w^{*T} x_i = x_i^T w^* \in \mathbb{R}$ so

$\Leftrightarrow$

$$\sum_{i=1}^{N} x_i (x_i^T w^*) = X^T y$$

$\Leftrightarrow$ by associativity and distributivity

$$\underbrace{\left(\sum_{i=1}^{N} x_i x_i^T \right)}_{= X^T X} w^* = X^T y$$

$$\Leftrightarrow X^T X w^* = X^T y \quad / \cdot (X^T X)^{-1}$$

$\Leftrightarrow$

$$w^* = (X^T X)^{-1} X^T y$$ is the optimal (and unique) solution. $\square$

b) $\eta :=$ step size. Then:

$$w_{t+1} = w_t - \eta \underbrace{\frac{1}{N} \left(\sum_{i=1}^{N} (w_t^T x_i) x_i - y_i x_i \right)}_{\nabla L(w)}$$ is the SGD update rule.

Equivalent:

$$w_{t+1} = w_t - \frac{\eta}{N} \sum_{i=1}^{N} (w_t^T x_i - y_i) x_i .$$

c, d in ipynb next.

(2) a) Since $L(w)$ is convex, we know that:

$$L(w_t) - L(w^*) \leq \langle \nabla L(w_t), w_t - w^* \rangle =$$

$$\nabla L(w_t) = \frac{x_t - x_{t+1}}{\eta} \Leftrightarrow \quad w_t = x_t - \eta P_t = x_{t+1} + \eta \nabla L(w_t) - \eta P_t$$

$$= \langle \nabla L(w_t), x_{t+1} - w^* + \eta(\nabla L(w_t) - P_t) \rangle =$$

$$= \langle \nabla L(w_t), x_{t+1} - w^* \rangle + \eta \langle \nabla L(w_t), \nabla L(w_t) - P_t \rangle =$$

$$= \langle \nabla L(w_t), x_{t+1} - w^* \rangle + \frac{\eta}{\eta} \langle x_t - x_{t+1}, \nabla L(w_t) - P_t \rangle =$$

$x_t = w_t + \eta P_t$ ... $= \frac{x_t - x_{t+1}}{\eta} - \frac{x_t - w_t}{\eta}$

$$= \langle \nabla L(w_t), x_{t+1} - w^* \rangle \pm \langle w_t - x_{t+1}, \nabla L(w_t) - P_t \rangle + \langle \eta P_t, \nabla L(w_t) - P_t \rangle =$$

$$= \langle \nabla L(w_t), x_{t+1} - w^* \rangle + \langle \nabla L(w) - P_t, w_t - x_{t+1} \rangle + \eta \langle P_t, \frac{x_t - x_{t+1} - x_t + w}{\eta} \rangle =$$

$$= \langle \nabla L(w_t), x_{t+1} - w^* \rangle + \langle P_t, w_t - x_{t+1} \rangle + \langle \nabla L(w) - P_t, w_t - x_{t+1} \rangle \square$$

which is the desired inequality.

b) First, note that: $\|a+b\|_2^2 = \|a\|_2^2 + \|b\|_2^2 + 2\langle a, b \rangle$. Using that:

$$\langle \nabla L(w_t), x_{t+1} - w^* \rangle = \langle \frac{x_t - x_{t+1}}{\eta}, x_{t+1} - w^* \rangle = \frac{1}{\eta} \langle x_t - x_{t+1}, x_{t+1} - w^* \rangle =$$

$$= \frac{1}{2\eta} \left(\|x_t - x_{t+1} + x_{t+1} - w^*\|_2^2 - \|x_t - x_{t+1}\|_2^2 - \|x_{t+1} - w^*\|_2^2 \right) =$$

$$= \frac{1}{2\eta} \left[\|x_t - w^*\|_2^2 - \|x_t - x_{t+1}\|_2^2 - \|x_{t+1} - w^*\|_2^2 \right]$$ which is what we wanted to show.

Next.

$$\langle p_t, w_t - x_{t+1} \rangle = \left\langle \frac{x_t - w_t}{\eta}, w_t - x_{t+1} \right\rangle = \frac{1}{\eta} \langle x_t - w_t, w_t - x_{t+1} \rangle =$$

$$= \frac{1}{2\eta} \left[\| x_t - w_t + w_t - x_{t+1} \|_2^2 - \| x_t - w_t \|_2^2 - \| w_t - x_{t+1} \|_2^2 \right] =$$

$$= \frac{1}{2\eta} \left[\| x_t - x_{t+1} \|_2^2 - \| x_t - w_t \|_2^2 - \| w_t - x_{t+1} \|_2^2 \right]$$ which we needed to show. Finally, let: $a = (\nabla L(w_t) - p_t)\eta$, $b = w_t - x_{t+1}$ find $\| a - b \|_2^2$:

$$\| \eta(\nabla L(w_t) - p_t) - (w_t - x_{t+1}) \|_2^2 = \| \eta(\nabla L(w_t) - p_t) \|_2^2 + \| w_t - x_{t+1} \|_2^2$$
$$- 2 \langle \eta(\nabla L(w_t) - p_t), w_t - x_{t+1} \rangle \geq 0$$

Then: $\Leftrightarrow$ because $\| \cdot \|_2^2 \geq 0$ for any $\| \cdot \|_2^2$.

$$\| \eta(\nabla L(w_t) - p_t) \|_2^2 + \| w_t - x_{t+1} \|_2^2 \geq 2\eta \langle \nabla L(w_t) - p_t, w_t - x_{t+1} \rangle$$

$$\Leftrightarrow$$

$$2\eta \langle \nabla L(w_t) - p_t, w_t - x_{t+1} \rangle \leq \eta^2 \| \nabla L(w_t) - p_t \|_2^2 + \| w_t - x_{t+1} \|_2^2$$

$$\Leftrightarrow$$

$$\langle \nabla L(w_t) - p_t, w_t - x_{t+1} \rangle \leq \frac{\eta}{2} \| \nabla L(w_t) - p_t \|_2^2 + \frac{1}{2\eta} \| w_t - x_{t+1} \|_2^2,$$

which we wanted to show. This shows all 3. □

c) First, combining the result of a) and b), we have:

$$L(w_t) - L(w^*) \leq \frac{1}{2\eta} \left[\| x_t - w^* \|^2 - \| x_{t+1} - w^* \|^2 + \| x_t - x_{t+1} \|^2 - \| x_t - w_t \|^2 \right]$$ (≤ 0, remaining inequality)
$$- \| w_t - x_{t+1} \|^2 + \| w_t - x_{t+1} \|^2 \Big] + \frac{\eta}{2} \| \nabla L(w_t) - p_t \|^2 \leq \frac{1}{2\eta} \left[\| x_t - w^* \|^2 - \| x_{t+1} - w^* \|^2 \right]$$
$$+ \frac{\eta}{2} \| \nabla L(w_t) - p_t \|^2.$$ Now let there be some T, then: (≤ 0)

$$T \left(\frac{1}{T} \sum_{t=1}^{T} L(w_t) - L(w^*) \right) \leq \frac{1}{2\eta} \left[\| x_1 - w^* \|^2 - \| x_2 - w^* \|^2 + \| x_2 - w^* \|^2 - \| x_{T+1} - w^* \|^2 \right] +$$
$$+ \frac{\eta}{2} \sum_{t=1}^{T} \| \nabla L(w_t) - p_t \|^2 \overset{\text{since } L \text{ is convex, s.o. } \sum_{t=1}^{T} \frac{1}{T} L(w_t) \geq L\left(\sum_{t=1}^{T} \frac{1}{T} w_t \right), \text{ let } \hat{w} = \sum_{t=1}^{T} \frac{1}{T} w_t}{\Rightarrow} L(\hat{w}) - L(w^*) \leq \frac{1}{2\eta T} D^2 + \frac{\eta}{2T} \sum_{t=1}^{T} \| \nabla L(w_t) - p_t \|^2$$

$\| x_1 - w^* \|^2 - \| x_{T+1} - w^* \| \leq D^2$ □

(3) a) $L(w) = w(w^2/3 - \bar{w}^2)$, $w \in \mathbb{R}$, $w \in [-\sqrt{3}\bar{w}, \sqrt{3}\bar{w}]$.

Convex:

$L(w) = \frac{w^3}{3} - \bar{w}^2 w \Rightarrow \nabla_w L(w) = w^2 - \bar{w}^2$, $\nabla_w^2 L(w) = 2w$.

Now, let $w = 0 \Rightarrow \nabla L(w) = -\bar{w}^2 < 0$ for any $\bar{w} > 0$, so not convex. If $\bar{w} = 0$, $w \in [0,0] \Leftrightarrow \{0\}$, then $\forall w \in \{0\}$, $\nabla L(w) = 0$ if $\bar{w} = 0$, so $\nabla L(w) = 0 \geq 0$ so $L(w)$ is convex. So:

- $L(w)$ is not convex on $[-\sqrt{3}\bar{w}, \sqrt{3}\bar{w}]$ for $\bar{w} \in \mathbb{R} \setminus \{0\}$
- $L(w)$ is convex on $[-\sqrt{3}\bar{w}, \sqrt{3}\bar{w}]$ for $\bar{w} = 0$

G-Lipschitz: Let: $x, y \in [-\sqrt{3}\bar{w}, \sqrt{3}\bar{w}]$:

$$|L(x) - L(y)| = \left|\frac{x^3}{3} - x\bar{w}^2 - \frac{y^3}{3} + y\bar{w}^2\right| = \left|\frac{1}{3}(x^2+xy+y^2)(x-y) - \bar{w}^2(x-y)\right|$$

$$= \left|\frac{1}{3}(\underbrace{x^2+xy+y^2}_{\geq 0}) - \bar{w}^2\right| |x-y| \underset{\substack{x=\sqrt{3}\bar{w} \\ y=\sqrt{3}\bar{w}}}{\leq} 2\bar{w}^2 |x-y|$$

, set $G = 2\bar{w}^2$, or $G \in \mathbb{R}^+$ if $\bar{w} = 0$, then $L(w)$ is G-Lipschitz

L-Smooth: Let $x, y \in [-\sqrt{3}\bar{w}, \sqrt{3}\bar{w}]$, :

$$\|\nabla L(x) - \nabla L(y)\| = |x^2 - \bar{w}^2 - y^2 + \bar{w}^2| = |x^2 - y^2| = |x+y||x-y| \leq$$

$\leq 2\sqrt{3}\bar{w}|x-y|$, set $L = 2\sqrt{3}\bar{w}$, or $L = 0$ for $\bar{w} = 0$, then $L(w)$ is L-smooth.

So: $L(w)$ is → not convex; → G-Lipschitz; → L-Smooth.

b) First, suppose $\bar{w} = 0 \Rightarrow w^* = 0$. Let $\bar{w} \in \mathbb{R} \setminus \{0\}$. Let $w = 0$, $L(0) = 0$, so $L(w^*) \leq L(0) = 0$. Then: $\nabla L(w') = 0 \Rightarrow w'^2 - \bar{w}^2 = 0$ $\Rightarrow w' = |\bar{w}|$ or $w' = -|\bar{w}|$. $L(-|\bar{w}|) = -|\bar{w}|\left(\frac{\bar{w}^2}{3} - \bar{w}^2\right) = -|\bar{w}| \cdot \left(-\frac{2}{3}\bar{w}^2\right) = \frac{2}{3}|\bar{w}|\bar{w}^2$ > 0, so $L(-|\bar{w}|) > 0$, so $w = -|\bar{w}|$ is not the minimizer. Next: Q

$L(|\bar{w}|) = |\bar{w}|\left(\frac{\bar{w}^2}{3} - \bar{w}^2\right) = -\frac{2}{3}|\bar{w}|\bar{w}^2 < 0$, and finally $\nabla^2 L(|\bar{w}|) = -2|\bar{w}| > 0$. So since $\nabla L(|\bar{w}|) = 0$ and $\nabla^2 L(|\bar{w}|) > 0$ and $-|\bar{w}|$ is not a minimizer, and $|\bar{w}|$ is a local minimum, and there are no other optimums apart from $-|\bar{w}|$, $|\bar{w}|$ is the global minimizer of $L(w)$ on $[-\sqrt{3}\bar{w}, \sqrt{3}\bar{w}]$.

c) In code

d) Given the update:

$w_2 = w_1 - \eta \nabla L(w_2) = w_1 - \eta\left(w_2^2 - \bar{w}^2\right) \Leftrightarrow \eta w_2^2 + w_2 - (w_1 + \eta\bar{w}^2) = 0$

This is quadratic in w_2: solve for w_2:

$$w_2 = \frac{-1 \pm \sqrt{1 + 4\eta(w_1 + \eta\bar{w}^2)}}{2\eta}, \text{ and we need } 1 + 4\eta(w_1 + \eta\bar{w}^2) \geq 0$$

$$\Rightarrow w_1 + \eta\bar{w}^2 \geq -\frac{1}{4\eta} \Leftrightarrow w_1 \geq -\left(\frac{1}{4\eta} + \eta\bar{w}^2\right). \text{ so for } w_1 \geq -\left(\frac{1}{4\eta} + \eta\bar{w}^2\right),$$

$w_2 = \frac{-1 \pm \sqrt{1 + 4\eta(w_1 + \eta\bar{w}^2)}}{2\eta}$ has a closed form solution □.

e) $\lim_{\eta\to\infty} w_2 = \lim_{\eta\to\infty}\left(\underbrace{\frac{1}{2\eta}}_{\to 0} \pm \sqrt{\underbrace{\frac{1}{4\eta^2}}_{\to 0} + \underbrace{\frac{w_1}{\eta}}_{\to 0} + \bar{w}^2}\right) = \pm|\bar{w}|$.

The feasible range: $w_1 \geq -\left(\frac{1}{4\eta} + \eta\bar{w}^2\right) \Rightarrow$ 2 cases:

$\bar{w} = 0$: $\lim_{\eta\to\infty} -\frac{1}{4\eta} + 0 = 0 \Rightarrow w_1 \geq 0$ when $\bar{w} = 0$

$\bar{w} \neq 0$: $\lim_{\eta\to\infty} -\frac{1}{4\eta} - \eta\bar{w}^2 = -\infty$, so $w_1 \in \mathbb{R}$ as $\eta \to \infty$ for $\bar{w} \neq 0$

(since $\forall x \in \mathbb{R}, x > \infty$)

(5) Let u be sampled from a unit ball in $\mathbb{R}$.

By the hint:

$$\mathbb{E}\left[p^u(w)\right] = \mathbb{E}\left[\frac{L(w+\mu u)}{\mu} u\right] \quad (\text{I use } \mathbb{E}_u = \mathbb{E} \text{ notation})$$

Now consider:

$$p^u(w) - g^u(w) = \frac{L(w+\mu u)}{\mu} u - \frac{L(w+\mu u) - L(w)}{\mu} u =$$

$$= \frac{L(w)}{\mu} u.$$

So their difference is $\frac{L(w)}{\mu} u$. Then:

$$\mathbb{E}\left[p^u(w) - g^u(w)\right] = \mathbb{E}\left[\frac{L(w)}{\mu} u\right] \overset{\text{by the linearity of } \mathbb{E}}{\Leftrightarrow} \mathbb{E}\left[p^u(w)\right] - \mathbb{E}\left[g^u(w)\right] = \frac{L(w)}{\mu}\mathbb{E}[u]$$

Since $\mathbb{E}$ is over u, so $\frac{L(w)}{\mu}$ is const w.r.p. to u. Now, if $u \sim \text{Uniform}[-1, 1]$ (i.e. sampled uniformly over the unit ball in $\mathbb{R}$), $\mathbb{E}[u] = \mathbb{E}_u[u] = 0$. So:

$$\mathbb{E}\left[p^u(w)\right] - \mathbb{E}\left[g^u(w)\right] = 0$$

$$\mathbb{E}\left[p^u(w)\right] \overset{(=)}{=} \mathbb{E}\left[g^u(w)\right]$$

And since $g^u(w)$ already was an unbiased estimator of the gradient of $L_\mu(w)$, and $\mathbb{E}[p^u(w)] = \mathbb{E}[g^u(w)]$ (u is drawn from the unit ball in $\mathbb{R}$), then $p^u(w)$ is also an unbiased estimator of $L_\mu(w)$. □